

JULIETA D. RUBIS

AI Engineer | Generative AI | LLMs | RAG | Machine Learning | AWS

Email: julietarubis@gmail.com

LinkedIn: <https://www.linkedin.com/in/julietarubis/>

GitHub: <https://github.com/julietarubis>

Portfolio: <https://julietarubis.github.io/portfolio/>

Hugging Face: <https://huggingface.co/spaces/julietarubis/health-chatbot>

Medium: <https://medium.com/@julietarubis>

PROFESSIONAL SUMMARY

AI Engineer with an MSc in Artificial Intelligence & Machine Learning and hands-on experience developing Generative AI applications, Retrieval-Augmented Generation (RAG) systems, predictive analytics solutions, and deep learning models. Experienced in building LLM-powered applications using AWS Bedrock, LangChain, and Hugging Face, with expertise spanning anomaly detection, real-time data streaming, and machine learning deployment. Passionate about designing practical AI solutions that solve real-world business problems.

CORE SKILLS

Programming: Python, SQL, JavaScript

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, LSTM, Autoencoders

LLMs & Generative AI: LangChain, Hugging Face, OpenAI API, AWS Bedrock, FAISS, RAG

Data Engineering: Apache Spark, Kafka, ETL Pipelines, Pandas, NumPy

Cloud & MLOps: AWS (SageMaker, EC2, S3, Bedrock), Docker, Git

Frameworks: Flask, Dash, Streamlit

PROFESSIONAL EXPERIENCE

Machine Learning Engineer (Intern)

Orcawise – Dublin, Ireland

Jun 2024 – Dec 2024

- Developed and deployed an LLM-powered EU AI Act compliance chatbot using LangChain, FAISS, and AWS Bedrock, enabling semantic search across 1,000+ documents and reducing information retrieval time by 60%
- Fine-tuned LLMs (Mistral-7B, Claude) using Hugging Face and AWS Bedrock, improving response relevance and contextual accuracy by 35% for domain-specific legal queries
- Contributed to a healthcare bias detection and analysis system, supporting data preprocessing, model evaluation, and visualization of fairness metrics

Key Achievements:

- Reduced information retrieval time by 60%
 - Improved fairness evaluation efficiency by 40%
-

AI PROJECTS

AI-Powered Wind Turbine Anomaly Detection & Predictive Maintenance

MSc Dissertation | University of Limerick | 2025

- Developed an end-to-end predictive maintenance system using SCADA data from six wind turbines.
- Built and evaluated LSTM Autoencoder, Bi-LSTM, and Deep Autoencoder models for anomaly detection.
- Integrated Apache Kafka and Apache Spark Structured Streaming for near real-time fault detection.
- Developed a Streamlit dashboard to visualize anomaly alerts, risk indicators, and maintenance insights.

- Designed a scalable architecture combining deep learning, streaming analytics, and predictive maintenance.

Airline Passenger Satisfaction Prediction

Higher Diploma Academic Project | College of Computing Technology | 2023

- Analysed 129,000+ airline passenger records to identify drivers of customer satisfaction.
- Built and evaluated Logistic Regression, Decision Tree, Random Forest, LDA, and Gaussian Naive Bayes models.
- Achieved **96.3% prediction accuracy** using Random Forest.
- Identified key service improvements related to online booking, Wi-Fi, and boarding experiences.

EDUCATION

MSc Artificial Intelligence & Machine Learning

University of Limerick, Ireland | 2023 – 2026

Higher Diploma – Artificial Intelligence Applications

College of Computing Technology | 2022 – 2023

Certificate in Software Engineering

Technological University of the Shannon | 2022 – 2022

Higher Diploma in Digital Marketing

Dublin Business School | 2021 – 2022

QQI Level 5 in Healthcare Support

B&B Nursing, Ireland | 2020 – 2021

BSc Information Technology

Jose Rizal University | 2005 – 2009

Dissertation: Payroll System using PHP and MySQL

CERTIFICATIONS

- AWS AI & ML Scholarship Nanodegree – Udacity (2025)
- AWS AI Programming with Python Scholarship Nanodegree Program – Udacity (2024)
- Python for Data Science and AI – IBM (2024)

ADDITIONAL EXPERIENCE

Healthcare Assistant (Career Bridge Role)

Nursing Home – Ireland | 2020 – 2022

Beech Park Nursing Home – Ireland | Feb 2023 – July 2023

Ashley Lodge Nursing Home – Ireland | 2025 – Present

- Maintained high-accuracy digital records in clinical systems
- Delivered patient care in high-responsibility environments, developing strong attention to detail and operational discipline.

REFERENCES AVAILABLE ON REQUEST